

An IoT-Enabled System for Real-Time Fuel Monitoring, GPS Tracking, and Anomaly Detection in Trucks Using Non-Parametric Unsupervised Machine Learning

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Abstract

Fuel accountability in delivery-truck operations requires more than a fuel gauge: it requires synchronized vehicle telemetry, route context, resilient communication, and an anomaly model that distinguishes abnormal fuel loss from traffic, idling, refueling, sender noise, and driver variability. This paper presents an implemented IoT-enabled fleet-monitoring prototype developed with Hino Parañaque Subic GS Auto Inc. The system acquires OBD-II fuel, speed, GPS, trip-state, and alert data through an embedded HMI; transmits telemetry through LoRa, LTE/GSM, WiFi, and offline replay; stores records in a cloud backend; and detects fuel anomalies using non-parametric unsupervised learning. The production detector is Isolation Forest Model C, supported by Matrix Profile as a subsequence detector and by a data-mining layer for operating-regime interpretation. Fuel anomalies are modeled as context-conditioned deviations between observed and expected fuel-level change under speed, RPM, engine load, throttle, route, driver, and motion state. Field and controlled-injection evaluation showed 1.04% MAPE fuel accuracy, 5.52 s median GPS sampling, 90.21% GPS fix availability, 92.9% precision and 1.09% false-positive rate for Model C under leave-one-trip-out validation, and a 95.0% alert-resolution rate. The results support the use of context-aware telemetry and interpretable anomaly signatures for practical fleet fuel monitoring, while parked-depot siphoning remains a future-work boundary requiring a dedicated stationary-state detector.

Keywords— Fuel monitoring, GPS tracking, anomaly detection, Internet of Things, unsupervised machine learning, Isolation Forest, Matrix Profile, fleet management.

1 Introduction

Fuel cost, fuel theft, unreported leakage, poor route visibility, and delayed alert response remain persistent problems in delivery-fleet operations. Traditional approaches often treat fuel monitoring, GPS tracking, and driver alerts as separate functions, which limits their ability to explain whether a fuel drop is normal consumption, traffic-related idling, sender noise, refueling recovery, or

a genuine anomaly. Prior IoT fuel-monitoring and tracking systems show the value of connected sensors but often remain hardware-only or rule-driven, while machine-learning studies on heavy vehicles frequently use datasets detached from an integrated field prototype [1–4].

This study addresses that gap by implementing and evaluating a complete edge-to-cloud truck-monitoring system. The prototype combines OBD-II fuel and engine telemetry, GPS tracking, prioritized communication, cloud storage, dashboard visualization, HMI and mobile trip controls, and unsupervised anomaly detection. The machine-learning component is intentionally context-aware: it does not flag fuel loss only because the level dropped. Instead, it evaluates whether the observed fuel change is plausible under the vehicle’s current operating context.

The specific contributions of this paper are: (1) an integrated IoT fuel, GPS, and alerting prototype for truck fleet monitoring; (2) a context-conditioned anomaly model grounded in physical and operational constraints; (3) an Isolation Forest production detector supported by Matrix Profile and K-Means-based operating-regime discovery; and (4) an evaluation across fuel accuracy, communication reliability, GPS cadence, anomaly-detection metrics, and alert usability.

2 Related Work

IoT-based fleet-monitoring systems have used GPS, cloud dashboards, RFID, GSM, and sensor modules to improve vehicle visibility and operational response [5–7]. Fuel-specific systems have monitored tank level, theft alarms, and basic fuel events using resistance, ultrasonic, or capacitive sensing [8–10]. These systems demonstrate practical monitoring value, but many are limited by fragmented architecture, small-scale validation, or rule-based detection.

Machine-learning work on fuel and vehicle anomaly detection shows that abnormal consumption can be detected from multivariate telemetry. Mumcuoglu et al. [4], for example, showed that fuel-consumption anomalies can be classified in heavy-duty vehicles, while interpretable anomaly-detection literature emphasizes the importance of explaining why a point is anomalous

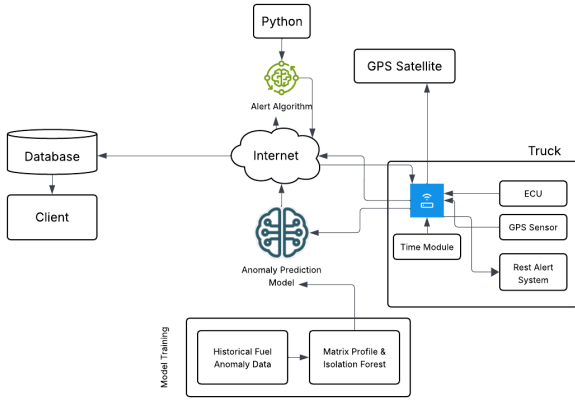


Figure 1: Conceptual diagram of the IoT-enabled fuel monitoring and anomaly-detection system.

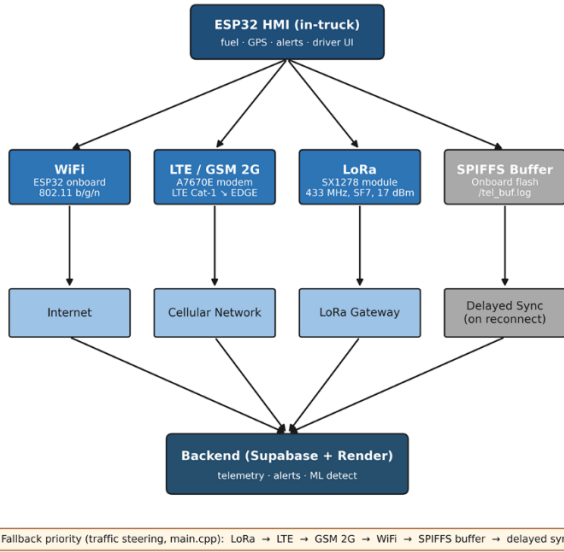


Figure 2: End-to-end edge-to-cloud system architecture.

rather than reporting only a score [11]. This study builds on those directions by deploying unsupervised models inside an implemented truck telemetry system and by tying every anomaly flag to physical or operational constraints.

3 System Architecture

Fig. 1 summarizes the conceptual system. The truck-side embedded HMI collects vehicle telemetry, displays local status, and allows driver trip interaction. The backend stores telemetry and trip state, evaluates anomaly and safety rules, and synchronizes the administrative dashboard, mobile companion interface, and HMI feedback path.

The implemented architecture in Fig. 2 follows a layered edge-to-cloud pattern. The edge layer performs OBD-II acquisition, GPS parsing, fuel filtering,

trip-state handling, HMI display, and communication-channel selection. The transport layer uses prioritized LoRa, LTE/GSM, WiFi, and SPIFFS offline replay so telemetry is not silently lost during degraded network conditions. The cloud layer persists data in Supabase, runs backend rules and anomaly inference, and exposes synchronized views to the dashboard, mobile app, and HMI poll-back path.

4 Methodology

4.1 Fuel Acquisition and Filtering

Fuel, speed, GPS, trip, and engine data are sampled by the HMI and transmitted as synchronized telemetry rows. Fuel readings are filtered before model inference to reduce sender noise and slosh. The filtering pipeline combines a median stage, exponential smoothing, and a dead-band publisher, allowing the system to suppress short tank-motion artifacts while preserving meaningful fuel trends.

Fuel accuracy was evaluated against pump-gas reference events and the known tank capacity. The primary accuracy metric was mean absolute percentage error (MAPE):

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{V_{\text{sensor},i} - V_{\text{ref},i}}{V_{\text{ref},i}} \right|. \quad (1)$$

4.2 Communication, GPS, and Backend Integration

Each telemetry row carries device-side timestamp information and channel metadata. Server-side reconciliation separates sampling cadence from backend-arrival delay. Communication was evaluated through per-channel latency, packet delivery, and offline-replay behavior. GPS was evaluated using inter-sample interval and fix availability because route context, distance, and motion state feed the anomaly detector and dashboard.

4.3 Fuel-Anomaly Model

A fuel anomaly is modeled as a telemetry row whose observed fuel-level change deviates from the expected fuel-level change under operating context. For row t ,

$$A(t) = \frac{|\Delta f_{\text{obs}}(t) - \mathbb{E}[\Delta f \mid c(t)]|}{\sigma(c(t))}, \quad (2)$$

where $c(t)$ includes speed, RPM, engine load, throttle, route, driver, and motion state. This makes the detector a context-conditioned fuel-deviation model rather than a simple sudden-drop detector.

The model is constrained by four physical constraints and four operational constraints. C1 conservation states that fuel decreases through consumption or unauthorized

removal and increases only during refueling. C2 bounded consumption limits plausible fuel loss by engine work and load. C3 sender non-linearity accounts for unstable tank-level bands. C4 slosh accounts for short fuel fluctuations caused by liquid movement. O1 traffic/idling, O2 route restrictions, O3 driver variability, and O4 refuel patterns encode Philippine fleet-operation context. These constraints are implemented through direction-aware rules, refuel gates, fuel-per-RPM and fuel-per-load features, expected-consumption residuals, idle and motion-state features, route-type features, driver fuel-economy deviation, and post-refuel flags.

4.4 Machine Learning and Data Mining

Isolation Forest Model C is the production row-level detector. Model A used baseline fuel, motion, engine, and refuel-window features; Model B added driving-zone features; and Model C added per-asset, per-driver, and route-type fuel-economy deviation features. Matrix Profile was retained as a supporting subsequence detector for gradual fuel-loss patterns, not as the main production classifier.

A data-mining layer supports interpretation. Constraint-violation profiling annotates each flagged row with signatures such as C2 bounded consumption, C4 slosh suppression, and O3 driver baseline deviation. K-Means clustering with $K = 4$ was used only to discover operating regimes in the controlled-injection window. The seven standardized clustering features were filtered fuel delta, speed, engine load, idle duration score, motion state, throttle, and normalized RPM. The four discovered regimes corresponded to high-speed cruise, mid-speed cruise, gradual-decline behavior, and raw-slosh artifact behavior.

4.5 Evaluation Protocol

Fuel accuracy, communication reliability, GPS cadence, anomaly detection, and alert usability were evaluated according to the study’s five objectives. SO4 anomaly-detection metrics were computed using leave-one-trip-out validation on the project-native partition so that each held-out trip was scored by models trained and calibrated on separate trips. Precision and false-positive rate (FPR) were computed as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{FPR} = \frac{FP}{FP + TN}. \quad (3)$$

5 Results and Discussion

Table 1 summarizes the main objective-level results. The fuel subsystem reached 1.04% MAPE and 3.00% maximum error across the 13–98% tank range, satisfying the $\pm 5\%$ target. GPS reached a 5.52 s median sampling interval and 90.21% fix availability. The communication subsystem demonstrated multi-channel delivery and

meaningful offline-buffer use, preventing telemetry loss during weak connectivity.

For anomaly detection, Model C was the strongest production detector because it considered fuel movement relative to driver, route, and operating context. It achieved 92.9% precision, 92.9% recall, and 1.09% FPR on the project-native partition. The result satisfies the 90% precision and 10% FPR requirements while preserving Model C as the deployable detector. The Hybrid Intersection produced 97.4% precision as a high-confidence alert channel, but Model C remains the primary production model because it provides the best row-level balance of detection and coverage.

The constraint-violation analysis makes the ML result auditable. In the controlled-injection analysis, O3 driver baseline deviation appeared in every flagged row, C2 bounded consumption appeared in 69.0% of flagged rows, and C4 slosh suppression appeared most strongly in sudden and rapid drops. The controlled-injection window had zero false positives, but this is scoped only to that window; the full leave-one-trip-out evaluation still reports the overall 1.09% FPR. K-Means archetype discovery further showed that labelled and predicted positives were concentrated in moving-vehicle operating regimes, while a raw-slosh artifact cluster contained zero labels and zero predictions, confirming that the filter cascade suppressed C3/C4 artifacts before inference.

The alert subsystem completed the operator workflow. Alerts were surfaced across fuel anomaly, overspeed, low fuel, rest, and maintenance classes, with synchronized visibility across the HMI, mobile app, and dashboard. The administrator System Usability Scale score of 81.9/100 indicates that the dashboard was not only functional but usable for routine monitoring and response.

6 Limitations and Future Work

The detector is strongest for in-trip and moving-vehicle anomalies because its context features depend on movement, speed, RPM, engine load, route progression, and driver-baseline deviation. Parked-depot siphoning, where the vehicle is stationary and engine-off, requires a separate stationary-state detector using long-duration fuel-level monitoring. Other limitations include the need for longer deployment with confirmed real-world theft events, broader vehicle and tank calibration, stronger modern cellular fallback, and more multi-user dashboard testing.

7 Conclusion

This paper presented an implemented IoT-enabled system for real-time truck fuel monitoring, GPS tracking, anomaly detection, and alert management. The prototype met the thesis objectives across fuel accuracy, com-

Table 1: Objective-Level Evaluation Summary

Objective	Target	Measured Result	Status
SO1 Fuel measurement	Fuel accuracy within $\pm 5\%$	1.04% MAPE; 3.00% maximum error; all validation events within tolerance	Met
SO2 Communication	Cloud-accessible real-time telemetry	LoRa, LTE/GSM, WiFi, and offline replay maintained telemetry flow under varied network conditions	Met
SO3 GPS tracking	5 s location logging and route context	5.52 s median sampling interval; 90.21% GPS fix availability	Met
SO4 ML anomaly detection	$\geq 90\%$ precision and $\leq 10\%$ FPR	Isolation Forest Model C: 92.9% precision, 92.9% recall, and 1.09% FPR under leave-one-trip-out evaluation	Met
SO5 Alerts and dashboard	Usable alert and monitoring workflow	100 alerts over 17.7 days; 95.0% resolution rate; administrator SUS score of 81.9/100	Met

munication, GPS cadence, machine-learning anomaly detection, and alert usability. The central technical finding is that fuel anomalies should be modeled as context-conditioned deviations, not as raw fuel drops alone. Isolation Forest Model C, supported by Matrix Profile and a constraint-aware data-mining layer, provided a practical and auditable production detector. Overall, the system demonstrates that embedded telemetry, resilient communication, contextual unsupervised learning, and human-readable alert explanations can support fuel accountability and proactive fleet supervision.

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